

PREDICTIVE ANALYTICS**Midterm Project***Individual Project Brief and Assessment Rubric*

Item	Specification
Assessment	Midterm Examination (UTS)
Mode	Individual project
Project options	Regression, classification, or time-series forecasting
Oral assessment	One-to-one interview during Meetings 7–8; maximum 7–8 minutes per student
Submission	Jupyter Notebook, dataset, and Executive Summary

Purpose

This project assesses the student's ability to complete an end-to-end predictive analytics workflow: define a relevant prediction problem, prepare real-world data, develop and compare models, evaluate performance correctly, interpret the results, and communicate evidence-based recommendations.

1. Project Task

Select one public dataset and complete one of the following predictive analytics projects. The dataset is freely chosen, if it satisfies all requirements in Section 3.

Project type	Prediction objective	Target example
Regression	Predict a continuous numerical value.	House price, demand, energy consumption
Classification	Predict a categorical class or outcome.	Customer churn, loan default, disease category
Time-series forecasting	Predict one or more future values while preserving time order.	Monthly sales, daily demand, inflation rate

2. Intended Learning Outcomes

- Formulate a clear predictive question, target variable, and unit of analysis.
- Assess data quality and perform justified data preparation and feature engineering.
- Build a valid baseline and the required predictive models for the selected task.
- Use an appropriate validation strategy and evaluation metrics without data leakage.
- Compare models, interpret feature importance, and select a final model based on evidence.
- Communicate conclusions, limitations, and actionable recommendations clearly.

3. Dataset Selection Requirements

Requirement	Minimum expectation
Public and verifiable	Use Kaggle, UCI Machine Learning Repository, World Bank, BPS, data.go.id, or another credible public/official source.
Source citation	Include dataset title, publisher/owner, access link, and access date in the notebook and Executive Summary.
Clear target	Define a meaningful response/target variable that matches the chosen project type.
Sufficient size	At least 500 observations and at least five potential predictor variables before feature selection.
Requires preparation	The data must contain realistic preparation needs; it must not be completely clean and model ready.
Authentic and ethical	No synthetic data and no personal, confidential, or sensitive information.
Computationally feasible	The complete notebook must run within a reasonable time in Google Colab.

Suggested data sources

<https://data.go.id/>

<https://www.bps.go.id/id>

<https://archive.ics.uci.edu/>

<https://www.kaggle.com/>

4. Required Submission Package

Submit one compressed folder through the course RISE using this naming convention:

StudentID_FullName_MidtermProject.zip

Required file	Expected format and content
Jupyter Notebook	. ipynb file containing the complete analysis and visible outputs.
Dataset	The exact data file(s) used by the notebook, in a portable format such as CSV or XLSX.
Executive Summary	A maximum of 2–3 pages in PDF or DOCX format.

5. Deliverable 1 — Jupyter Notebook

The notebook must be organized as a reproducible analytical report. Use Markdown cells to explain the reasoning before or after each major code section. A correct output without explanation is not sufficient.

No.	Required section	Minimum content
1	Project overview	Problem context, prediction objective, stakeholder/value, task type, target, and unit of analysis.
2	Dataset description and source	Source citation/link, dimensions, observation unit, target, predictors, and relevant data dictionary.
3	Data quality assessment	Data types, missing values, duplicates, invalid values, target distribution, and possible leakage.
4	Data preparation	Documented cleaning, type correction, missing-value handling, duplicate handling, and justified outlier treatment.
5	Exploratory data analysis	Focused univariate and multivariate analysis with readable tables/visuals that inform modeling.
6	Feature engineering	Task-relevant transformations, encoding, scaling, date/time features, or aggregation, with rationale.

5. Deliverable 1 — Jupyter Notebook (continued)

No.	Required section	Minimum content
7	Baseline model	A simple benchmark that establishes the minimum performance a trained model must beat.
8	Model training	Required models, correct split logic, fitted preprocessing, reproducibility controls, and concise rationale.
9	Model evaluation	Appropriate metrics, comparison table, diagnostics, and evidence-based final model selection.
10	Feature importance	Interpret global feature influence with a suitable method; explain limitations and avoid causal claims.
11	Conclusion and recommendations	Direct answer, key findings, practical recommendations, limitations, and next steps.

6. Minimum Modeling Requirements

Project type	Required minimum models	Recommended evaluation evidence
Regression	Baseline; linear regression; at least one tree-based or ensemble model.	MAE and RMSE; R^2 as supporting information; residual/error analysis.
Classification	Baseline; logistic regression; at least one tree-based or ensemble model.	Confusion matrix; precision, recall, F1-score; accuracy and/or ROC-AUC where appropriate.
Time series	Naive baseline; at least two forecasting approaches.	Chronological evaluation; MAE/RMSE and MAPE only when valid; forecast plot and error analysis.

7. Validation and Technical Requirements

- Use a validation strategy that matches the task. Random splitting may be used for suitable tabular data; time-series data must be split chronologically.
- Fit preprocessing only on training data. Do not use validation/test information to impute, scale, select features, or tune models.
- Use the same evaluation split(s) and primary metric when comparing candidate models.
- State the primary metric and justify why it matches the problem and the costs of errors.
- For imbalanced classification, report class-sensitive metrics and discuss the decision threshold when relevant.
- For time series, inspect trend, seasonality, time gaps, and forecast horizon; do not shuffle observations.
- Set random seeds where applicable and keep the workflow computationally reasonable.
- Feature importance indicates predictive contribution, not automatically causal influence.

8. Reproducibility Standard

The submitted notebook must run successfully using:

Restart Runtime → Run All

- No execution error, missing variable, hidden manual step, or dependency on a previous session.
- Use relative paths or a single clearly defined data path; do not use a personal local-computer path.
- Include all required imports and installation commands at the beginning, only when necessary.
- The included dataset must be the same version used to generate the reported results.
- Outputs, tables, plots, labels, and units must be readable and consistent.

9. Deliverable 2 — Executive Summary

Prepare a concise, decision-oriented summary of no more than 2–3 pages. It should be understandable to a non-technical stakeholder while remaining faithful to the analysis.

Required element	What to communicate
Problem background	Why the problem matters and who could use the prediction.
Dataset and target	Source, scope, unit of analysis, target, and important limitations.
Methods compared	Baseline and candidate models, with a short rationale.
Main results	Key metric values and the most important diagnostic evidence.
Selected model	Which model is recommended and why it offers the best trade-off.
Recommendations	Practical action or next analytical step supported by the evidence.
Limitations	Data, modeling, validation, interpretation, and generalization limits.

10. Individual Oral Interview

Each student will complete a one-to-one interview during Meetings 7–8. Maximum duration: 7–8 minutes. The purpose is to verify individual ownership, conceptual understanding, and the ability to interpret the project—not merely to present polished outputs.

- Be ready to open the notebook, locate relevant cells, and explain the code and outputs.
- Be ready to justify one preparation choice, one metric, and the final model selection.
- The lecturer may ask for interpretation, identification of leakage risk, or a small code change.
- A separate slide deck is not required unless announced by the lecturer.

12. Assessment Rubric — 100 Points

The project is graded using evidence in the submitted package and the individual interview. Full-credit evidence is described below.

Criterion	Pts	Evidence expected for full credit
1. Problem formulation and dataset suitability	10	Clear predictive question, relevant target/unit; public, cited dataset satisfies size, predictor, ethics, and feasibility requirements.
2. Data quality assessment and preparation	15	Systematic audit; justified treatment of missing values, duplicates, types, invalid values, and outliers; no leakage; reproducible steps.
3. Exploratory analysis and feature engineering	15	Focused, readable EDA informs decisions; meaningful features and correct transformations; explanations connect evidence to modeling.
4. Baseline and model development	15	Valid baseline and all required models; appropriate preprocessing and splitting; correct implementation and justified choices.
5. Evaluation and model selection	15	Suitable metrics; fair comparison; relevant diagnostics; final model chosen using validation evidence, not training performance alone.
6. Feature importance, conclusions, recommendations	10	Sound interpretation; no unsupported causal claims; direct conclusion, specific recommendations, and honest limitations.
7. Reproducibility and code quality	5	Restart Runtime → Run All succeeds; portable path; organized code; files included; clear Markdown and readable outputs.
8. Executive Summary	5	Complete, accurate, concise, stakeholder-friendly 2–3 page summary.
9. Individual interview and ownership	10	Accurate explanation; justifies decisions, interprets results, navigates notebook, and answers independently.
TOTAL	100	The total project score is converted to the 15% Midterm Assessment contribution.

How partial credit is determined

Performance level	General interpretation
Complete and convincing	Correct, fully documented, justified, reproducible, and interpreted accurately.
Mostly complete	Core work is correct, but explanation, evidence, diagnostics, or polish is incomplete.
Partially demonstrated	Some relevant work is present, but important errors, omissions, or weak justification limit confidence.
Not demonstrated	Required evidence is absent, unusable, substantially incorrect, or cannot be explained.

Important grading implications

- A notebook that does not complete Restart Runtime → Run All successfully cannot receive full reproducibility marks.
- Missing baselines, required model families, or valid evaluation design reduces the corresponding marks.
- Target leakage, train/test contamination, or inappropriate time-series splitting is a major methodological error.
- Unsupported causal statements, copied interpretations, or inability to explain work reduces interpretation and interview marks.
- Meeting the minimum model count alone does not guarantee a high score; reasoning, validation, interpretation, and reproducibility are assessed.

Submission file

1. File .ipynb from deliverable 1
2. File .pdf from deliverable 2
3. File .pdf (screenshot that you have upload file 1-2 to your own github- please attached your github account/username here)
4. Dataset or any other required attachment for reproducibility.

Submit files point 1-4 in **compressed .zip** and upload it to RISE.

Interview schedule

Important: Your submitted work will only be assessed if you attend the scheduled interview session. If you do not attend the interview, your entire submission will not be graded.

Senin 28 September 2026

No.	Waktu (WIB)	Nama Mahasiswa
1	20.00-20.08	Muhammad Abdullah
2	20.08-20.16	Zulfikar Ditya Antariksa
3	20.16-20.24	David Abadi Anugrah
4	20.24-20.32	Yeni Nur Arwanti
5	20.32-20.40	Fajar Dwiharjo
6	20.40-20.48	Ariel Sharon Ferdinandus
7	20.48-20.56	Zaky Hafiedz
8	20.56-21.04	Sayyid Abdul Aziz Haidar
9	21.04-21.12	Farisyah Aulia Hanum
10	21.12-21.20	Husni
11	21.20-21.28	Nadira Dika Kaivalya
12	21.28-21.36	Ilham Notohusodo
13	21.36-21.44	Tita Noviana

Selasa 29 September 2026

No.	Waktu (WIB)	Nama Mahasiswa
1	20.00-20.08	Suci Fransisca Sisilia R
2	20.08-20.16	Muhamad Helmi Ramadhan
3	20.16-20.24	Rafli Ramadhan
4	20.24-20.32	Randi Ranata
5	20.32-20.40	Emerald Alphante Reirezqi
6	20.40-20.48	Adlan Muammar Shiddiq
7	20.48-20.56	Taufiqurrahman
8	20.56-21.04	Riyo Rialdy Urda

No.	Waktu (WIB)	Nama Mahasiswa
9	21.04-21.12	Wildan Rizky Wijaya
10	21.12-21.20	Titanio Yudista
11	21.20-21.28	Zainuddin
12	21.28-21.36	Pebry Zauhary
13	21.36-21.44	Zaki Khairi Ziwar